

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester I Examination, 2024-25 (NEP)

CHEMISTRY**Paper Code: CHEMN-11 : Basic Concepts of Chemistry — I**

Time: 2:30 Hours

Maximum Marks: 70

REF NO. CECNF/N/2024-25/29

*Instructions: Answer **five** questions in all including Question No. 1 which is compulsory. All questions carry equal marks (14 marks each).*

Question 1. Short Answer Questions (Compulsory): [2 × 7 = 14]

- (a) (i) What is the physical significance of + and – signs in atomic wave function orbitals? [1]
- (ii) What are the essential conditions for the acceptability of wave functions obtained by solving the Schrödinger equation? [2]
- (b) What are the major consequences of the inert pair effect in *p*-block elements? [2]
- (c) Obtain the mathematical expressions for van der Waals constants *a* and *b* in terms of critical constants *P_c*, *V_c*, *T_c*, and *R*. [3]
- (d) State the First Law of Thermodynamics and write its fundamental limitations. [3]
- (e) Prove that the work done during isothermal reversible expansion of an ideal gas is greater than that in adiabatic reversible expansion. [3]

Question 2.

- (a) Calculate the effective nuclear charge (*Z*^{*}) on the outermost valence electron of ²⁹Cu using Slater's rules. [2]
- (b) What are eigenfunctions and eigenvalues? Give two examples for each concept. [2]
- (c) What are the solubility trends of alkali metal fluorides and iodides in water? Discuss the reasons based on lattice energy versus hydration energy. [4]
- (d) What are the major limitations of the Born-Landé equation for lattice energy calculation? [2]
- (e) (i) Calculate the standard reduction potential for Fe³⁺/Fe. Given *E*^o(Fe³⁺/Fe²⁺) = +0.77 V and *E*^o(Fe²⁺/Fe) = –0.47 V. [2]
- (ii) Will Fe²⁺ disproportionate into Fe³⁺ and Fe in acidic solution? Support your answer with thermodynamic calculation. [2]

Question 3.

- (a) Derive the 3D time-independent Schrödinger wave equation assuming the electron behaves as a 3D standing de Broglie wave. [4]
 - (b) Qualitatively discuss the periodic trends of first ionization energies for the elements of the 2nd and 3rd periods, explaining anomalies at B, O, Al, and S. [4]
 - (c) Discuss the dependence of lattice energy on ionic size and explain the consequences of ionic size on lattice energy and hydration enthalpy. [3]
 - (d) Calculate the electronegativity of carbon using Pauling's scale if Δ (extra bond energy for C-H bond) is 24.3 kJ mol^{-1} . Explain the concept of extra bond energy. [3]
-

Question 4.

- (a) State the fundamental postulates of the kinetic molecular theory of gases, providing experimental evidence for each postulate. [4]
 - (b) Explain critical temperature (T_c), critical pressure (P_c), and critical volume (V_c) with suitable examples. How is V_c determined experimentally? [5]
 - (c) State the Law of Corresponding States and derive the van der Waals reduced equation of state. [5]
-

Question 5.

- (a) The number of drops of a liquid and water were observed to be 66 and 30 respectively using a stalagmometer. Calculate the surface tension of the liquid if the surface tension of water is 72.75 dyne/cm , and the masses of liquid and water for the same volume are 10.750 g and 11.250 g respectively. [5]
 - (b) What is the Law of Rational Indices? Why is it necessary to convert Weiss indices into Miller indices? Write the steps involved in calculating Miller indices. [4]
 - (c) Draw neat diagrams and explain the crystal structures of graphite and diamond, detailing hybridization, bonding, and physical properties. [5]
-

Question 6.

- (a) Using the fundamental thermodynamic relation:

$$\mu_{JT} = \frac{1}{C_p} \left[T \left(\frac{\partial V}{\partial T} \right)_P - V \right]$$

show under what conditions the Joule-Thomson coefficient (μ_{JT}) is positive (cooling effect) for a real gas. [5]

- (b) Why is cooling of real gases observed during Joule-Thomson expansion? Show that Joule-Thomson expansion is an isenthalpic process ($H = \text{constant}$). [5]
-

- (c) What is the physical significance of entropy? “The entropy of the universe is constantly increasing.” Justify this statement using the Second Law of Thermodynamics. [4]
-

Question 7.

- (a) Prove the Maxwell thermodynamic relation:

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial P}{\partial T}\right)_V$$

[5]

- (b) “The total heat absorbed by a system cannot be completely converted into useful work.” Explain this statement based on the Carnot cycle efficiency. [5]
- (c) Show that $-\Delta G = W_{\text{net}}$ (useful net work). Under what thermodynamic condition does the change in Gibbs free energy (ΔG) equal the change in Helmholtz work function (ΔA)? [4]